

SAEHUN SEAN OH

Sr. Data & Platform Engineer · Streaming / OTT

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SUMMARY

Data and platform engineer with 10+ years shipping consumer video products, now owning the dbt / Airflow / BigQuery analytics platform ODK Media runs its business on. Set the modeling standards for a new dbt pipeline replacing a legacy Luigi chain, cut one daily validation from a 186 GB to a 1.5 GB scan per run (~99%), and automated partner revenue-data ingestion that had been running on manual dashboard downloads. Ten years of Android, Roku, and Chromecast engineering sits behind the data work, so the models are designed by someone who knows how the events are emitted.

SKILLS

Languages: Kotlin, Java, TypeScript, Python, SQL

Data & Infrastructure: dbt, Apache Airflow, Google BigQuery, Apache Kafka, AWS (Lambda), Google Analytics 4 (GA4)

Mobile & Connected TV (CTV): Android (Jetpack Compose, ExoPlayer), Roku (SceneGraph, BrightScript, BrighterScript), Chromecast (CAF Web Receiver)

Streaming & Video: HTTP Live Streaming (HLS), client-side ad insertion (CSAI), server-side ad insertion (SSAI), Mux, video on demand (VOD), over-the-top (OTT)

Tools & Practice: AI-assisted development (Claude Code, ChatGPT), GitHub Actions, Firebase

Spoken Languages: English, Korean (bilingual)

EXPERIENCE

ODK Media *Streaming / OTT · OnDemandKorea (ODK/ODKR) & Amasian*

2020 – Present

Sr. Data & Platform Engineer

2026 – Present

- Own ODK's analytics data platform (dbt / Airflow / BigQuery); set the data-modeling standards and conventions for a new analytics pipeline and designed the migration of core viewership models off a legacy Kafka / Luigi / BigQuery chain and its layered ad-hoc scheduled queries onto version-controlled, tested dbt pipelines, now beginning rollout; designing the revenue-share settlement system to land on it.
- Built a daily 3-way viewership-hours validation reconciling Kafka, GA4, and Mux, cutting the daily scan from **186 GB to 1.5 GB (~99%)** by rebuilding its source as a partitioned dbt model in place of the unpartitioned legacy ingest table.
- Rebuilt the revenue-share settlement workflow: automated the Amagi partner data transfer into S3 with validation surfaced through an interactive Slack bot (Bolt on AWS Lambda) that lets non-engineers investigate and acknowledge data-quality issues via slash commands, replacing manual dashboard downloads and row-by-row checks; moved its BigQuery scheduled queries onto service-account ownership with backfill, ending failures tied to one person's credentials and fixing duplicated settlement rows.
- Own audience-data strategy for addressable advertising: delivered content-affinity and language pools, unauthenticated-audience sizing, and the Publica ad-impression analytics feed powering an internal business intelligence tool; researching and designing how non-exclusive inventory-partner and owned-and-operated data can be enriched through identity federation providers (ID5, UID 2.0, LiveRamp).
- Drove adoption of AI-assisted development across ODK's US engineering teams as the earliest advocate for Claude and ChatGPT; co-authored the shared CLAUDE.md configurations and reusable Claude Code skills now used beyond the data team, and coached engineers in other units on subagent-driven workflows and token-efficient tooling.

Chromecast Receiver Development — Self-Initiated Project (concurrent)

2026 – Present

- Shipped two production Chromecast receivers from scratch in TypeScript (ODKR and Amasian), deployed on Vercel, self-teaching Google's CAF Web Receiver platform with no prior Cast or web background.
- Implemented server-side ad-insertion behavior (pre-roll on resume, ad-boundary seek snapping, subtitle clearing) and optimized for legacy low-memory devices through 720p capping, heap monitoring, and memory-leak fixes.
- Wrote the sender-side Cast integration in both Android apps.

Roku Engineer

2024 – 2026

- Primary Roku engineer for both apps (BrighterScript / SceneGraph); delivered a full deep-linking implementation across episode, movie, series, and live content and carried it through multiple Roku certification processes.
- Shipped EPG (electronic program guide) promo banners with QR-code panels and “More to Watch” live recommendations with pagination; standardized Mux metadata across both apps.
- Built a shared, versioned Roku data library consumed as a Git submodule with automated sync and linting tooling, eliminating duplicated code across the two apps.
- Automated the Roku build and release pipeline with GitHub Actions, replacing manual release-candidate handovers that were slow and required sharing signing credentials.

Project Coordinator / Technical Liaison, CTV Apps (concurrent with Roku Engineer)

2024 – 2025

- Served as point of contact between ODK’s US office and an offshore team in Vietnam building smart-TV apps across LG webOS, Samsung Tizen, Vizio, Amazon Fire TV, and Android TV.
- Managed requirements, scope, schedules, and releases across five CTV platforms and multiple time zones simultaneously — translating product needs into engineering work and reviewing deliverables — while continuing hands-on Roku development.

Android Engineer

2020 – 2024

- Built the first-generation OnDemandKorea Android app, then led the ground-up v2 Kotlin / Jetpack Compose rewrite, driving a major player refactor and standardizing live and VOD playback event handling.
- Sustained **99% crash-free sessions across 2M+ monthly view hours** via a Firebase + Slack live-monitoring setup.
- Integrated Mux video-performance analytics, wiring player state (fullscreen, channel switching, error handling) into the analytics layer, plus third-party ad SDKs including PubMatic OpenWrap header bidding.
- Built a OneSignal notification system driving **18K+ app visits per month**; integrated Google Play and Roku Billing with backend purchase validation.

Smule

2016 – 2020

Senior Android Application Developer

- Maintained a **99% crash-free rate across ~13M installs** through close Firebase Crashlytics monitoring; led core projects and code reviews without sacrificing delivery velocity.
- Led the codebase transition from Java to Kotlin and set the team’s coding conventions.
- Built reusable, scalable custom Views and dynamic content lists (RecyclerView / Adapter / ViewHolder); integrated OkHttp/Retrofit, Picasso, ExoPlayer, MoPub ads, and Facebook / Twitter / Snapchat / YouTube sharing APIs.

DiscreteUnits

Jan 2016 – May 2016

Android Application Developer - Contract

- Built a restaurant ordering app (menu browsing, tab open/close, in-app ordering); integrated Parse, Facebook, Clover, and Stripe for user credentials, catalog, and payments.

MaXentric

2015 – 2016

Android Application Developer

- Built the MaXJacket front-end app replacing Motorola radio hardware with VoIP alongside P25; used GPS / Google Maps and Android sensors; optimized and debugged the app to **raise crash-free users by 40%**.

EDUCATION

B.S. Computer Science · University of California, San Diego